Case Study: AI-Powered Transport Invoice Reconciliation Workflow

Public portfolio version - screenshots and business data are anonymized/redacted.

Built an end-to-end workflow that ingests invoice emails, parses documents with AI/programmatic fallbacks, matches invoice lines against business data, supports human review, and records outbound sent-mail lifecycle.

Business problem

Transport invoice processing required repeated mailbox checks, PDF handling, rate matching, exception review, and outbound forwarding to finance. The workflow needed to reduce manual handling while keeping exceptions reviewable and auditable.

Solution summary

Step	Stage	Purpose
1	Mailbox ingestion	Synchronize transport invoice emails and store attachments.
2	Classification and parsing	Use program parsing first, with AI fallback for unparseable or unknown documents.
3	Persistence and readiness	Persist invoice data, attempt downstream sync, then dispatch rate-fill only when required data is ready.
4	AI-assisted rate fill	Use an AI job to select matching quote lines and transport rates, followed by backend cross-checking.
5	Review and send lifecycle	Separate review status from outbound send status; send approved invoices and persist sent-log records.

Key engineering highlights

• End-to-end mailbox-to-review-to-send workflow rather than a single isolated parsing feature.
• Readiness-based dispatch to avoid doomed rate-fill runs when order, quote, or levy data is missing.
• AI fallback path for document classification and parsing while keeping program parsing as the primary path.
• Review status and send status separated so outbound mail progress is not mixed into review semantics.
• Backend recomputation of comparison summary and send decisions instead of trusting frontend-provided flags.
• Best-effort downstream sync that records misses without rolling back invoice persistence.

Workflow architecture

1	Azure Mailbox Ingest
2	Attachment Stored
3	Document Classification
4	Program Parse / AI Fallback
5	Invoice / Quote / Levy Persistence
6	Best-effort Order Data Sync
7	Readiness Evaluation
8	Transport Rate Fill Job
9	Cross-check and Comparison
10	Review Sync
11	Approved Outbound Send
12	Azure Sent Log

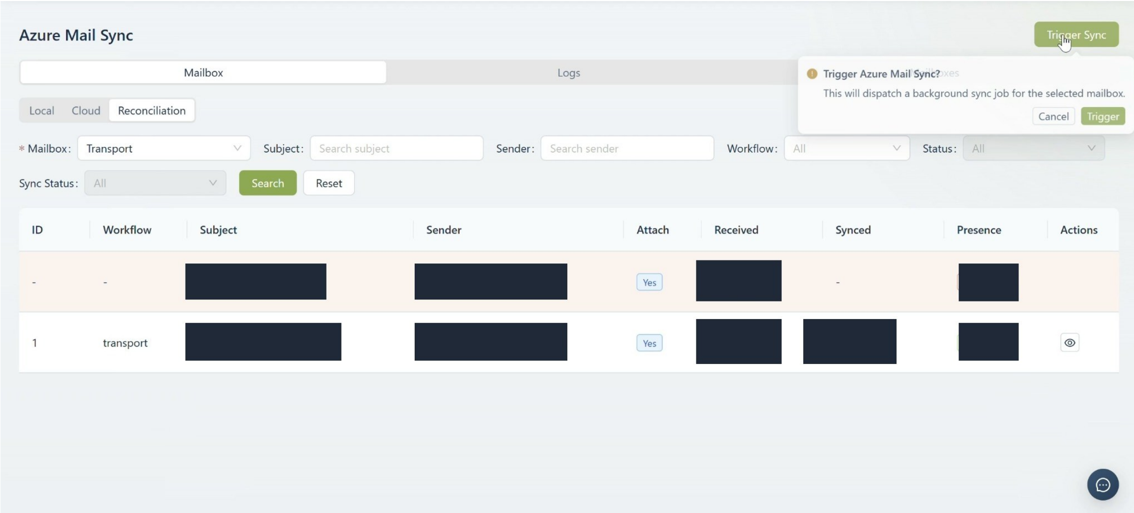
Design notes

The baseline treats transport_rate_fill as an invoice-driven downstream step. It is dispatched only after order data sync has been attempted and readiness checks confirm enough data for a meaningful run. When readiness is blocked, the workflow writes an ignored job with a reason instead of enqueueing a doomed AI run.

Review acceptance comes from corrected comparison outcomes. The backend rebuilds the comparison summary and recomputes whether abnormal results remain before approving and attempting outbound send.

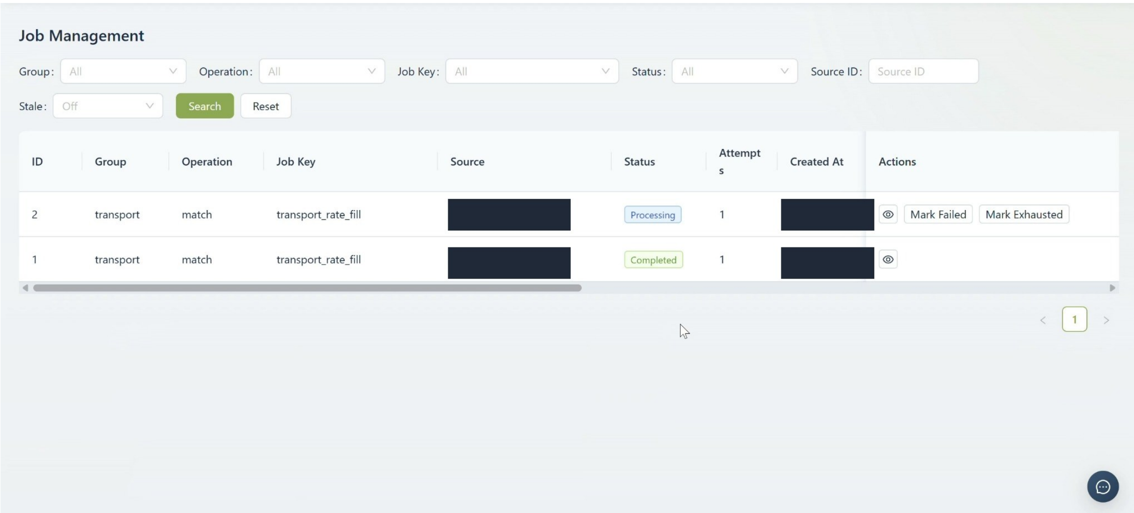
Screenshots: workflow in action

1. Mailbox ingestion / Azure Mail Sync



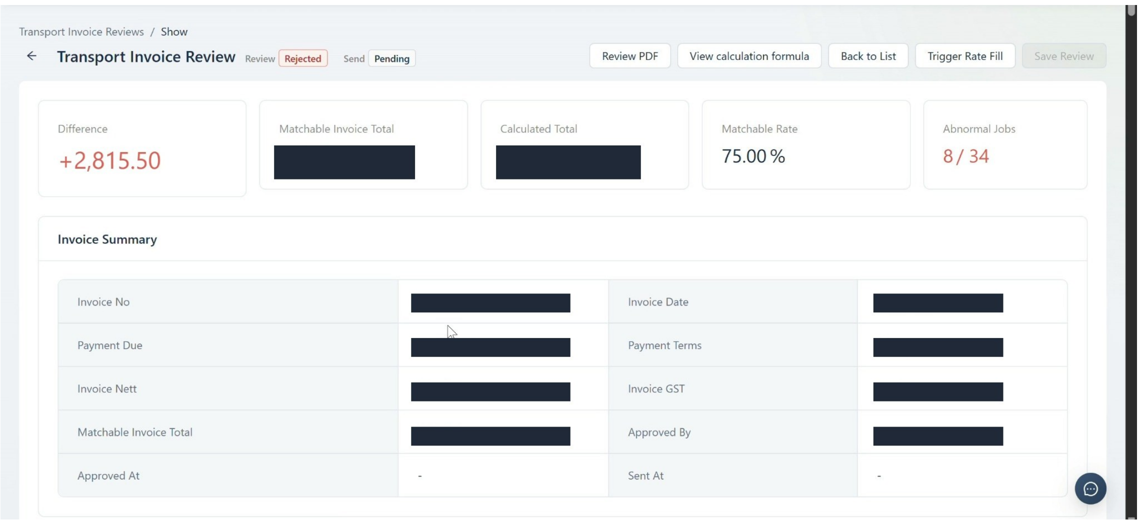
Invoice emails are synchronized from an Azure mailbox into the transport workflow. Sensitive subjects, senders, timestamps, and internal details are redacted.

2. Background AI job monitoring



Transport rate-fill jobs are tracked through observable processing and completed states. This demonstrates AI workflow orchestration rather than ad hoc prompt execution.

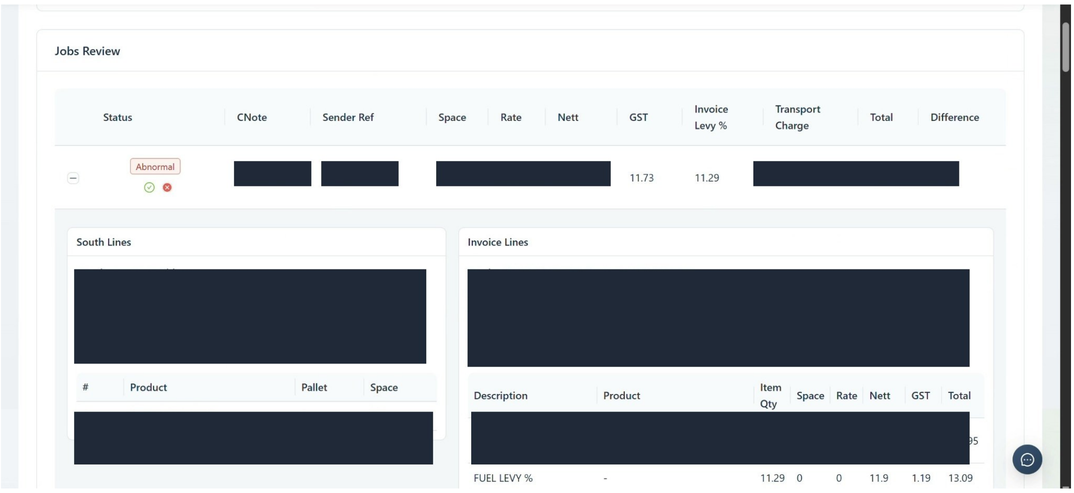
3. Invoice review summary



Parsed invoice totals are compared against calculated transport values, with review and send states shown separately for operator visibility.

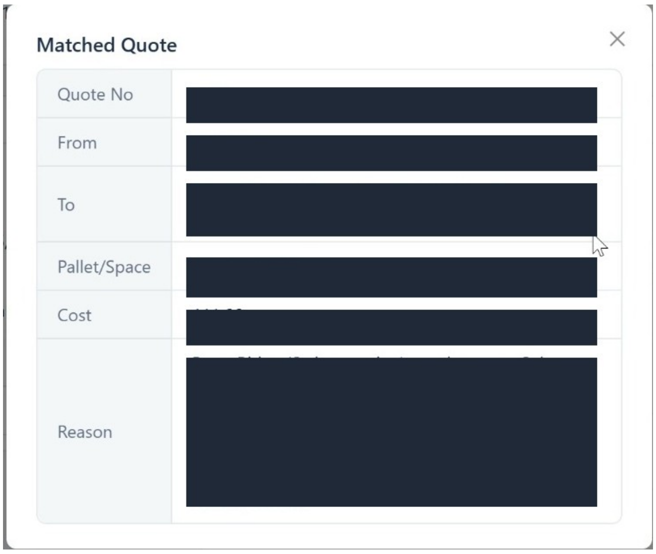
Screenshots: review details

4. Line-level comparison



The review page supports line-level comparison between business data and invoice lines. Business values are redacted while workflow structure remains visible.

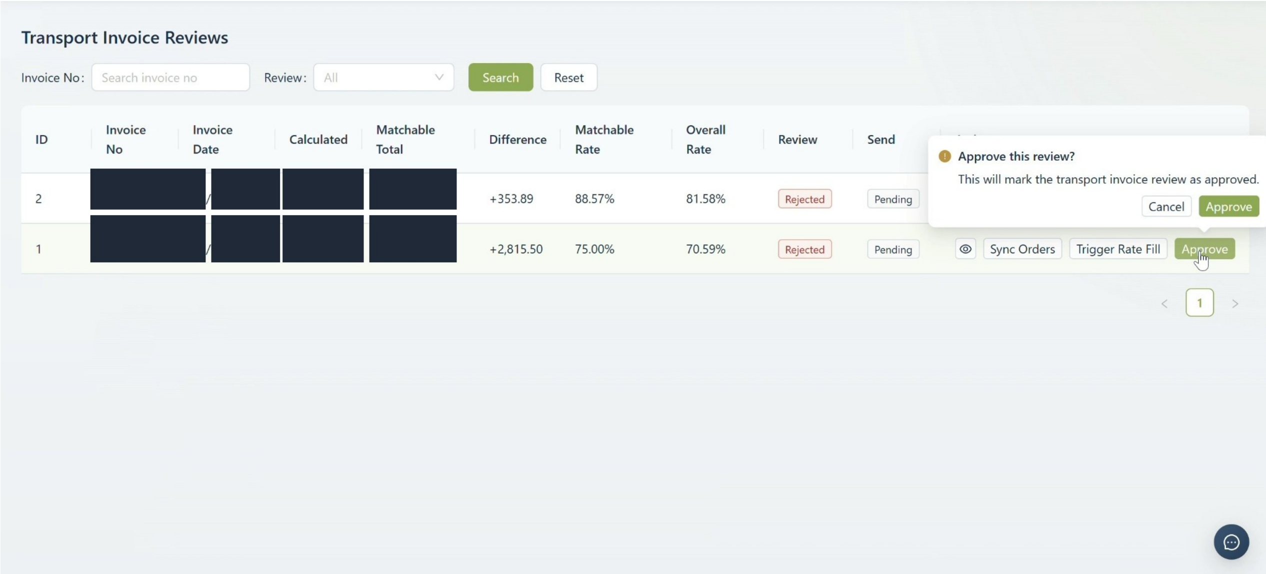
5. Matching explanation



The matching modal displays how a quote/rate decision was selected. Sensitive locations, costs, and customer details are redacted.

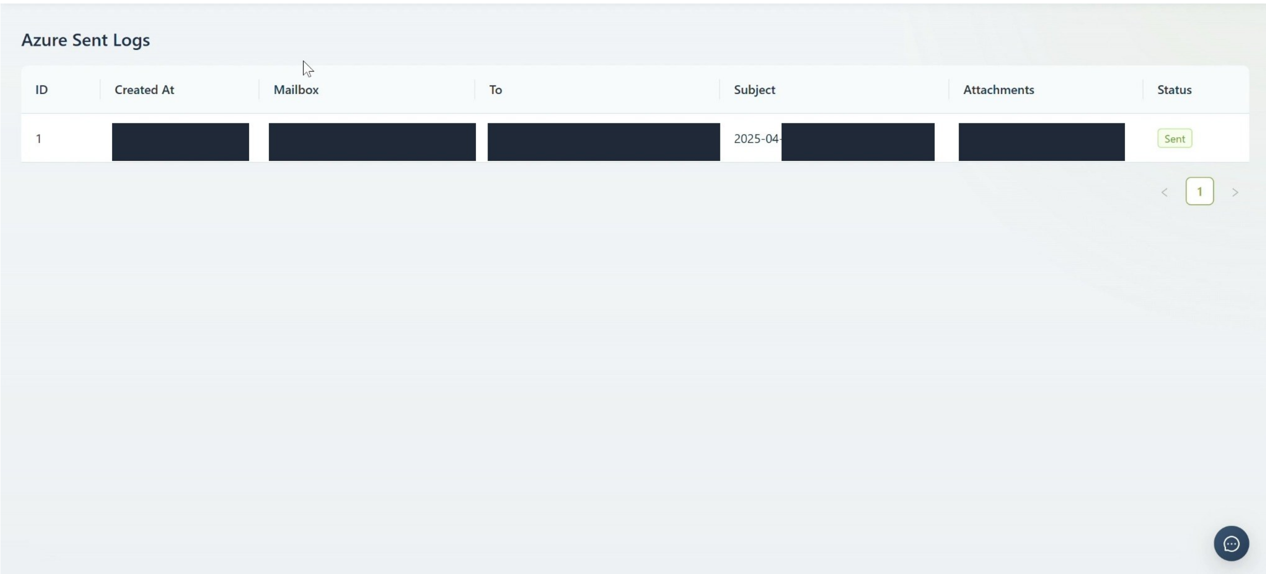
Screenshots: review approval and outbound send

6. Manual review and approval



Operators can trigger sync/rate-fill actions and approve reviews after comparison outcomes are resolved.

7. Outbound send lifecycle



Approved invoices are sent through the Azure mailbox flow and recorded in sent-log records with send status.

Confidentiality and portfolio note

This public case study is based on an anonymized enterprise automation workflow. Screenshots are redacted and simplified for portfolio use. Business data, client information, internal identifiers, production URLs, email addresses, invoice numbers, and financial values are not disclosed.

Suggested LinkedIn Featured title

AI-Powered Invoice Reconciliation Workflow on Azure - Case Study

Suggested LinkedIn description

Designed and delivered an end-to-end workflow for transport invoice reconciliation, covering Azure mailbox ingestion, AI-assisted parsing, readiness-based job dispatch, review workflow, and outbound sent-log tracking. Professional reference from Barden Farms available upon request.